

PUBLICATIONS

18 TOTAL PAPERS	17 PEER-REVIEWED	1 IN REVISION	6 FIRST / CO-FIRST
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SELECTED PUBLICATIONS · FIRST / CO-FIRST AUTHOR

- 2026** Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. IF 27.7 [in revision](#)
- First author (co-first). **Design** — four matched cohorts (healthy control, psoriasis, rheumatoid arthritis, ankylosing spondylitis), 16 donors each, n = 64; memory CD4⁺ T enrichment by negative selection, 4-plex pooling with SNP demultiplexing so donor and sequencing run are not confounded; 5' scRNA-seq with paired TCR. 251,055 memory CD4⁺ T cells. **Analysis** — built custom UMI-aware, BAM-level code (Python) resolving PDE4D short- vs long-isoform identity from 5' reads, which standard count matrices cannot separate; consensus NMF (k = 14 → 12 programs); scRNA + scTCR + snATAC + STAT3 ChIP-seq integration; Milo KNN-graph differential abundance; condition classifier on Th1.17 subset composition — arthritis AUC 0.780 (95% CI 0.665–0.896), healthy control 0.771, psoriasis 0.806. **Generalisation** — the PDE4D-high Th1.17 state was conserved across the SCAID atlas (702 samples, 23 immune diseases) and in independent SLE and arthritis synovial-fluid datasets. **Causal test** — short-isoform overexpression (PDE4D1/2/6) raised the IFN-γ⁺IL-17A⁺ fraction while the long isoform PDE4D5 did not; the PDE4D-selective inhibitor zatolmilast lowered it.
- 2025** **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. IF 2.9 [Download PDF](#)
- First author (co-first). Built the complete brain-immune single-cell analysis path (whole-brain harvest → enzymatic digestion → Percoll enrichment → FACS → 10x → analysis); identified exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium (↑ MHC-I and interferon-stimulated genes) at the blood-CSF barrier. 11,587 cells, 10 immune populations.
- 2025** **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. IF 12.6 [Download PDF](#)
- First author (co-first). Single-cell and functional dissection of dysfunctional skin-homing (CLA⁺) regulatory T cells in atopic dermatitis; OX40 (TNFRSF4) up-regulated alongside impaired suppressive function.
- 2024** Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. IF 3.9 [Download PDF](#)
- First author (co-first). Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques; longitudinal immune monitoring.
- 2022** Youngkyoung Lim*, **Seong-Jun Kang***, Bum Keun Cho, Hyun Je Kim, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. IF 9.2 [Download PDF](#)
- First author (co-first). Ran the GeoMx DSP workflow end to end — assay setup and instrument operation, immunofluorescence staining (S100B / CD3ε / nuclei), fluorescence-guided ROI selection and imaging, barcode collection, then the image and spatial-transcriptomic analysis. **Design** — one stage-IIC melanoma arising from a 4 cm congenital melanocytic nevus, segmented into four regions by invasion depth (M1–M4, 300 → 4,200 μm Breslow), four whole-transcriptome ROIs per region, 16 ROIs total, 18,677 genes. 2,067 differentially expressed genes (11.07%) at P < 0.0005; graded loss of HLA class I/II with depth, and B2M loss confirmed by immunohistochemistry.

- 2019** Jong Min Kim, Jun Seop Shin, Byoung Hoon Min, **Seong-Jun Kang***, Il Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung Soo Hwang, Jongwon Ha, Chung Gyu Park. "JAK3 inhibitor–based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. IF 2.2 [Download PDF](#)
First author (co-first). JAK3–inhibitor–based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

- 2026** Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh–Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides–like Atopic Dermatitis Represents a Th22–dominant Inflammatory Endotype." *Allergy*. IF 12.6
Performed the scRNA–seq & scTCR–seq analysis (scVI cross–dataset integration with public MF/AD cohorts, InferCNV genomic–identity inference, scTCR clonality). Showed mfAD is a reactive, oligoclonal Th22 endotype rather than a malignancy, responsive to JAK1 inhibition. 248,881 cells.
- 2024** Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim–Schulze, Chung–Gyu Park, Hyun Je Kim, Tae–Gyun Kim. "High–dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin–trafficking property." *eBioMedicine*. IF 11.1 [Download PDF](#)
Co-author. High–dimensional immune profiling of regulatory T cells in psoriasis, revealing an impaired skin–trafficking phenotype.
- 2024** Youngkyoung Lim*, Beom Keun Cho*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan–Sik Park, Je–Ho Mun, Chong Hyun Won, Chung–Gyu Park. "Spatial transcriptomic analysis of tumour–immune cell interactions in melanoma arising from congenital melanocytic nevus." *JEADV*. IF 9.2 [Download PDF](#)
Co-author. Spatial transcriptomic (GeoMx) analysis of tumour–immune cell interactions in nevus–arising melanoma.
- 2023** Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung–Gyu Park. "Mesenchymal stem cell–derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification." *Experimental & Molecular Medicine*. IF 12.8
Co-author. MSC–derived extracellular vesicles suppress Th17 cells by destabilizing RORγt via post–translational modification.
- 2022** Sangho Lee, Han–Teo Lee, Young Ah Kim, Il–Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung–Gyu Park, Kyungho Choi, Hong–Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin–mediated T–cell activation." *Experimental & Molecular Medicine*. IF 12.8
Co-author. An optimized CABIN1–derived core peptide inhibits calcineurin–mediated T–cell activation.
- 2021** So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung–Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." *Current Diabetes Reports*. IF 4.2 [Download PDF](#)
Co-author. Review of novel immunomodulatory strategies for porcine islet xenotransplantation.
- 2020** Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung–Gyu Park. "Mesenchymal stem cell–derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." *Immunology Letters*. IF 4.4
Co-author. MSC–derived exosomes suppress T–cell proliferation through p27kip1/Cdk2–mediated cell–cycle arrest.
- 2019** Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun–Seop Shin, Bongi Kim, Jong Min Kim, Jung–Sik Kim, Il–Hee Yoon, Byoung–Hoon Min, **Seong-Jun Kang**, Yong–Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung–Gyu Park. "Bioinformatic analysis of peripheral blood RNA–sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig–to–non–human primate islet xenotransplantation." *Scientific Reports*. IF 4.6
Co-author. Bioinformatic analysis of peripheral–blood RNA–seq to detect causes of late graft loss in pig–to–NHP islet xenotransplantation.
- 2018** Byoung–Hoon Min, Jun–Seop Shin, Jong–Min Kim, **Seong-Jun Kang**, Hyun–Je Kim, Il–Hee Yoon, Su–Kyoung Park, Ji–won Choi, Min–Suk Lee, Chung–Gyu Park. "Delayed revascularization of islets after transplantation by IL–6 blockade in pig–to–non–human primate islet xenotransplantation model." *Xenotransplantation*. IF 3.9
Co-author. IL–6 blockade delays islet revascularization in a pig–to–NHP islet xenotransplantation model.

- 2018** Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation*. IF 3.9
Co-author. Pre-clinical pig-to-NHP islet xenotransplantation with anti-CD40 (2C10R4)-based immunosuppression.
- 2018** Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." *Transplantation*. IF 6.2
Co-author. Pre-clinical islet allo-transplantation using a JAK inhibitor in place of tacrolimus in cynomolgus macaques.
- 2018** Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." *Biochemical and Biophysical Research Communications*. IF 3.1
Co-author. Engineered EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation.

MANUSCRIPTS IN PREPARATION

- 2026** Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." in preparation
Co-author (immunology). Performed the plasma proteomics analysis (Olink immune & neurology panels across the largest MSMDS patient cohort to date, n = 12 patients vs 6 controls), revealing broad down-regulation of lymphoid regulators (LTβR, BAFF) and leukocyte-adhesion molecules, with dimensionality reduction separating patients from controls; complemented by flow-cytometry immunophenotyping of lymph-node leukocytes (germinal-center B cells GL7+B220+) linking impaired lymphatic contractility to germinal-center loss and systemic immune dysregulation.

RESEARCH GRANTS & FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER – ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
NAVER Corporation	Fundamental therapeutic strategy for immune-mediated diseases using engineered Tregs (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)
Korea Drug Development Fund (KDDF)	Evaluation of drug efficacy and discovery of pharmacodynamic biomarkers of a new anti-CD40 antibody lead compound (PB101) for the treatment of pemphigus vulgaris	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Korea Drug Development Fund (KDDF)	Lead selection and optimization of a therapeutic anti-CD40 antibody for the treatment of multiple sclerosis	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
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Ministry of Science and ICT (MSIT) · Bio & Medical Technology Development Program	Establishing a meta-platform for discovering molecular targets by building a multi-layered immune-disease single-cell map (A9-22-1-05; SCAID, Single-cell Atlas for Immunological Disease)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.

■ TECHNICAL SKILLS

Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; Seurat, Scanpy, Squidpy, Monocle3, CellChat/NicheNet . Integration & generative models (as tools): scVI / scANVI , batch integration, consensus NMF , embedding geometry. Custom UMI-aware, BAM-level isoform analysis code (Python) ; two end-to-end analysis paths built from scratch. Graph/statistical methods: Milo KNN-graph DA, GLIPH , Shannon-entropy clonality, InferCNV, weighted-LMM/EMM. Spatial: GeoMx DSP (Whole Transcriptome Atlas), Visium spatial transcriptomics, MACSima cyclic multiplexed immunofluorescence imaging. Proteomics: Olink targeted plasma proteomics (immune & neurology panels).
Machine learning / deep learning / AI	Python ML/DL: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers; VAE/GAN/GNN exposure. Multimodal DL integration (imaging + transcriptomic + clinical); vision-language transformers. AlphaFold2/3, ProteinMPNN, DiffDock; MolFormer. Linux servers · R/RStudio · Git-based reproducible analysis.
Experimental (domain ground truth)	Primary mammalian cell culture: human T-cell isolation, expansion and functional assay; mesenchymal stem cell differentiation and culture; monocyte/dendritic-cell differentiation. 10× 5' / 3' scRNA-seq + scTCR-seq library generation; 10× Chromium Connect automated library preparation; cell/nucleus isolation. High-dimensional flow cytometry (up to 24 markers); multiplex panel design; 10-color 6-way FACS. In situ multiplexed assay operation: GeoMx DSP run end to end — slide preparation, oligonucleotide-barcoded probe hybridisation, immunofluorescence staining, fluorescence-guided ROI selection, instrument operation, UV cleavage and collection; MACSima cyclic multiplexed immunofluorescence (iterative stain-image-erase rounds). Microscopy: spinning-disk confocal (Nikon CSU-W1 SORA) , point-scanning confocal (Leica SP8) , two-photon; fluorescence microscopy; IHC/IF (frozen & FFPE) with quantitative image analysis. Brain/CNS immune profiling (perfusion, digestion, Percoll, CD45 ⁺ FACS). NHP CNS tissue (rhesus, cynomolgus, marmoset); EAE/MS & islet-transplant disease models.

■ RESEARCH INTERESTS

- **Reading immune cell state from incomplete single-cell data.** Cell-state calling under ambiguity, isoform and transcript resolution from short reads, and embedding representations of cells, with calibrated confidence.
- **Multi-modal integration of molecular identity.** Transcriptome, chromatin, spatial, and proteome into cell-state-resolved models.
- **Neuro-immune interactions in neurodegeneration.** T cells, microglia, and the blood-CSF barrier in Alzheimer's and related disease; ligand-receptor cross-talk and clonal/repertoire structure.
- **Building and checking computational analysis paths.** Custom callers, reproducible end-to-end analysis code, and comparison against domain ground truth.
- **Atlas-scale single-cell analysis across diseases and species.** Large multi-donor atlases (SCAID), cross-disease meta-analysis, and NHP-to-human translation.

■ AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

- 2024** **Antagonistic Anti-CD40 Humanized Antibody.** Patent No. 1020240085972 (registered 2024-07-01).
- 2019** **Immunosuppression Composition Comprising JAK Inhibitor.** Patent No. 1020143490000 (registered 2019-08-20).

AWARDS

- 2024** **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team)
- 2023** **Grand Prize**, SBA Smart Workathon (Work Process Innovation)
- 2017** **Best Poster Award**, American Transplant Congress, Chicago · **Best Presentation Award**, Korean Association of Immunologists

SCHOLARSHIPS

- 2014-2020** NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship

■ SELECTED PRESENTATIONS

- Poster** KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

■ REFERENCES

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